Secure User Authentication Scheme Using Multimodal Biometrics

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Abstract

Authentication systems increasingly require stronger protection against credential theft, replay, and biometric spoofing while remaining practical to deploy. Although multimodal biometrics can improve authentication robustness, conventional approaches often require separate acquisition of multiple traits, increasing hardware requirements, cost, and user interaction. This paper presents a deployable multimodal authentication system that derives face, voice, and a dynamic challenge response factor from a single real time video recording. During login, the user records a video while responding to a server issued CAPTCHA. Face verification and speaker verification are performed using learned biometric embeddings and cosine similarity based decisions, while the spoken response is transcribed and validated against the session specific challenge. The three factors are evaluated concurrently and combined using a conjunctive decision rule, granting access only when all verification factors succeed.

The system therefore enables single sensor acquisition of multiple authentication factors without requiring dedicated biometric hardware for each modality. The per session challenge also binds the login attempt to a fresh server issued value, limiting straightforward replay of previously captured recordings. For privacy, uploaded biometric recordings are reduced to templates and are not retained as persistent media and the resulting templates are encrypted, while per user key material is maintained separately from the template store. The implemented system was evaluated through deployment oriented processing measurements, demonstrating a practical concurrent authentication workflow while identifying template extraction and model loading as the main contributors to authentication latency. The available corpus is insufficient for formal False Acceptance Rate (FAR), False Rejection Rate (FRR), or Equal Error Rate (EER) estimation. The proposed architecture provides a foundation for practical, privacy conscious multimodal authentication using commonly available camera and microphone hardware.

Keywords: Authentication, Identity, Passwords, Biometrics, 2FA, MFA, Multimodal biometrics

Nomenclature: FAR, False Acceptance Rate; FRR, False Rejection Rate; EER, Equal Error Rate; ROC, Receiver Operating Characteristic; DET, Detection Error Tradeoff; AUC, Area Under the Curve; PAD, Presentation Attack Detection; STT, Speech to Text; TTL, Time to Live; MITM, Man in the Middle.

1 Introduction

1.1 Authentication

Authentication is the process of verifying the identity of an entity before granting access to a system, service, or resource. In user authentication, the user presents one or more credentials that are verified by the system. Authentication factors are commonly classified as knowledge, possession, and inherence factors. Knowledge factors include passwords and other information known to the user, possession factors include physical devices or tokens such as mobile phones and smart cards, and inherence factors are based on biometric characteristics such as fingerprints, facial features, iris patterns, or voice.

This work focuses on user to device authentication, in which a user is authenticated before being granted access to a protected resource. Traditional authentication systems commonly rely on passwords or other knowledge based credentials. However, such credentials may be exposed, guessed, stolen, or reused. Biometric authentication instead uses characteristics associated with the individual and can therefore provide an additional basis for identity verification. Combining multiple biometric traits can further reduce reliance on a single characteristic.

The proposed work uses face and voice as biometric factors and combines them with a dynamic challenge response mechanism. The user submits a real time recording from which the biometric and challenge response factors are derived, and access is granted only when all required verification decisions succeed. This motivates the multimodal authentication architecture presented in the following sections.

1.2 Cataloguing of Authentication Schemes

Authentication schemes can be broadly classified according to the number and type of credentials used for verification, as illustrated in Fig. 1. Single factor authentication relies on one credential, whereas two factor and multi factor authentication combine multiple factors to provide additional layers of verification. Authentication factors are commonly drawn from knowledge, possession, and inherence categories.

Biometric authentication belongs to the inherence category and uses physiological or behavioural characteristics to verify identity. Examples include fingerprints, facial features, iris patterns, voice, and gait. Conventional unimodal biometric systems rely on a single trait and may be affected by variations in sensing conditions, data quality, or limited discriminative information. Multimodal biometric systems

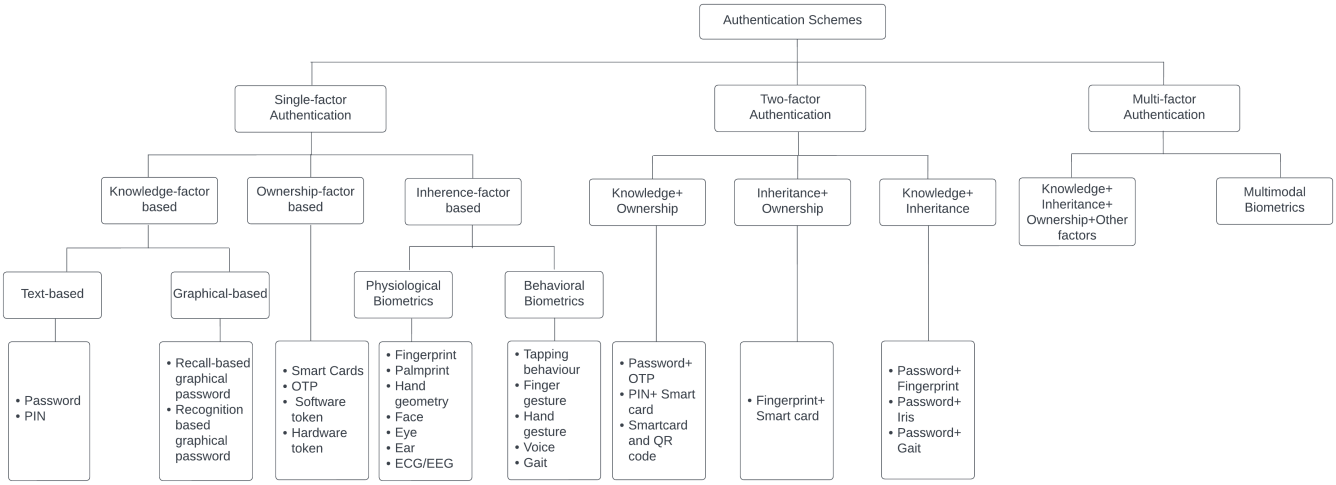


Fig. 1: Cataloguing of Authentication Schemes

address these limitations by combining multiple biometric traits, providing the motivation for the face and voice based authentication architecture proposed in this work.

This paper focuses on multimodal biometric authentication rather than providing a detailed taxonomy of password, possession based, or graphical authentication schemes.

1.3 Research problem and motivation

Researchers are trying to find a more secure and robust authentication mechanism to combat security threats and prevent the system from cyber attacks. Single factor authentication can be vulnerable to compromise of its single authentication factor, motivating the use of multiple complementary factors. Passwords may be exposed, guessed, stolen, or reused through attacks such as guessing and brute force attacks. MFA and passwordless authentication is gaining popularity and becoming the current trend. In the previous subsection, the benefits of biometric authentication are discussed and how multimodal biometric authentication can be a viable solution. Multimodal biometrics, which involves more than one biometric trait, is a secure and effective authentication mode. All the schemes based on multimodal biometrics involve some problems like

- interacting with different devices to provide biometric identity
- The extra cost involved in purchasing devices

So, many multimodal biometric systems use separate acquisition modalities or dedicated sensing hardware, which can increase deployment complexity and cost. The proposed scheme instead combines parallel acquisition of biometrics from a commodity camera and microphone with binding of that acquisition to a fresh server issued challenge.

1.4 Major Contributions

Below are the main contributions of this work. First, we present a framework for passwordless multimodal biometric authentication that combines facial and speaker recognition with a dynamic challenge response mechanism within a single authentication session. To prevent successful authentication through the compromise of a single authentication factor, the proposed framework employs a strict AND based fusion strategy, requiring the simultaneous verification of the facial biometric, speaker biometric, and session specific challenge response. Second, we introduce a server generated, unpredictable, and single use dynamic CAPTCHA as a challenge response mechanism to mitigate the reuse of previously recorded authentication media. Since each challenge is associated with a distinct authentication attempt, a response recorded during a previous session cannot satisfy the challenge issued in a subsequent session. Third, the proposed architecture reduces reliance on a strictly sequential authentication pipeline by concurrently evaluating the biometric and challenge response factors and applying a unified authentication decision after verification. Fourth, the framework adopts a privacy oriented biometric data management approach in which raw enrollment and authentication recordings are processed transiently and are not retained as permanent biometric records, while facial and speaker representations are extracted as templates and protected using authenticated encryption. Finally, the system incorporates purpose bound enrollment authorization and identity binding. Normal enrollment cannot overwrite an existing template, while authenticated re enrollment can replace it only after successful three factor verification. The rest of this paper is organized as follows. Section II reviews the related literature, and Section III presents the proposed multimodal biometric authentication system and its methodology. A comparison with recent authentication systems follows the proposed system discussion. Section IV reports the evaluation and results, Section V outlines future work, and Section VI concludes the paper.

2 Related Work

Biometric authentication has evolved from single trait verification towards multi factor and multimodal approaches that combine biometric information with additional security mechanisms. Marasco and Albanese [1] proposed FingerPIN, a multi factor authentication approach that combines

knowledge and inherence factors through a fingerprint based personal identification number (PIN), while an improved version of the scheme was subsequently introduced to strengthen security and usability [2]. Li and Hwang [3] proposed a remote biometric authentication scheme based on biometric verification, one way hashing, and smart card technology. Das [4] later proposed an improved scheme, however, it was shown to lack resistance against offline password guessing and privileged insider attacks and did not provide user anonymity. To address these limitations, Lee and Hsu [5] introduced an alternative approach based on extended chaotic maps.

Other studies have investigated the integration of biometric traits with cryptographic mechanisms and machine learning techniques. Jagadiswary and Saraswady [6] proposed an authentication scheme combining RSA asymmetric cryptography with biometric traits such as fingerprints, retina scans, and finger-vein patterns. Finger vein based authentication and identification have also received attention, including approaches focused on improving vein image quality and reducing noise during feature extraction [7], as well as convolutional neural network based biometric identification systems [8]. These studies demonstrate the applicability of cryptographic and learning based techniques for strengthening biometric authentication, however, they primarily focus on different biometric modalities and acquisition settings from those considered in the present work.

More recent research has increasingly explored multimodal authentication through different combinations of biometric traits and fusion strategies. Hammad et al. [9] proposed sequential and parallel multimodal authentication systems using ECG and fingerprint biometrics and employed CNN and Q-Gaussian multi support vector machine (QG-MSVM) based processing. Kim and Hong [10] combined teeth images and voice biometrics, where individual matching scores were integrated using weighted score level fusion. Ahamed et al. [11] proposed an AI based device level authentication model using ECG and PPG signals, while [12] investigated the combination of fingerprint, face, and iris biometrics using different machine learning based fusion and classification approaches. Collectively, these studies demonstrate the benefits of combining multiple biometric traits, however, they differ in terms of the biometric modalities considered, acquisition requirements, fusion strategies, and security mechanisms.

The proposed work differs from these approaches by integrating facial verification, speaker verification, and a session specific spoken challenge within a unified passwordless authentication framework. Rather than relying solely on biometric score fusion, the system independently verifies the facial biometric, speaker biometric, and challenge response and applies a strict decision level AND rule for the final authentication decision. The dynamic challenge additionally mitigates the reuse of previously recorded authentication media by requiring a response to a fresh challenge associated with the current authentication attempt. The proposed architecture also incorporates concurrent factor evaluation, privacy oriented template only storage with authenticated encryption, and an authorized enrollment mechanism. A more detailed comparison with recent end to end biometric authentication systems follows the proposed system discussion and is summarized in Table 2.

3 Proposed System and Methodology

The proposed scheme is a multimodal biometric authentication system that combines face verification, speaker verification, and a dynamic challenge-response factor. All three factors are derived from a single real time video recording, allowing facial and voice information to be acquired simultaneously while the user speaks a server issued challenge. Authentication succeeds only when the face, voice, and challenge response decisions are all positive under a conjunctive decision rule.

During registration, the user's biometric representations are extracted from the submitted recording and retained as encrypted templates rather than as raw biometric media. Each template is protected using an independent per user encryption key, while the corresponding protected key material is maintained separately in the key vault database. During login, a newly submitted recording is used to derive face and voice representations and to validate the spoken challenge. The biometric representations are compared with the enrolled references, while the transcribed response is verified against the challenge issued for the authentication attempt.

The dynamic challenge provides freshness to the authentication attempt and helps prevent simple reuse of previously recorded authentication responses. The three verification results are combined using AND logic, therefore, access is granted only when all three factors are successfully verified.

The proposed authentication scheme consists of three phases:

- Registration Phase
- Login and Challenge Issuance Phase
- Authentication Phase

3.1 Registration Phase

The registration phase begins when the user submits the required account information. After successful account creation, the server issues a short lived enrolment authorization token that must accompany the registration video submission. The client then records the video while speaking the displayed registration challenge, which is used only to elicit speech for speaker template extraction and is not verified as an authentication factor, challenge response verification is performed during login. The recording is processed to extract the facial and voice biometric representations required for authentication, after which the raw recording is discarded. The resulting biometric template is encrypted using an independently generated per user key, while the corresponding encrypted key is maintained separately in the key vault database. The biometric template is then bound to the enrolled account. If enrolment or template extraction fails, the registration process is rejected and the incomplete enrolment state is removed. The complete registration workflow is specified in Algorithm 1.

The enrollment authorization uses a 10 minute Time to Live (TTL). Measured server side enrollment processing ranged from 4.6 to 12.6 seconds, while the complete registration workflow ranged from 7.4 to 14.3 seconds. The ten

minute interval therefore accommodates recording, upload, processing, and several user retries after a failed recording, while limiting the exposure window of a bearer credential that authorizes biometric enrollment for a named account. Table 1 distinguishes the implemented challenge, grant, cleanup, and session lifetimes. Their ordering is deliberate: the 15 minute unenrolled account grace exceeds the 10 minute enrollment grant lifetime, which exceeds the 5 minute login challenge lifetime. Consequently, the janitor does not delete an account while a valid enrollment grant may still be used.

Username and email address uniqueness are enforced, with email compared through the keyed normalized index `idx`. Biometric uniqueness across accounts is not enforced and is out of scope because a global duplicate check would require cross user template comparison and conflict with per user key isolation. An abandoned account that has never completed enrollment contains no biometric template and may be released by the existing janitor after its grace period. An enrolled account is not silently replaced through normal enrollment: Algorithm 1 rejects an existing template. Replacement is permitted only through the authenticated re enrollment lifecycle described in Section 3.9.

Algorithm 1 User Registration / Enrolment

Require: identifier u; email address m; optional avatar; live recording capability
Ensure: an encrypted biometric template, or rejection
Phase 1: Account Creation
Client submits (u, m, avatar)
Validate u and m; reject invalid input
$\texttt{idx} \leftarrow \mathcal{H}_{k_{\text{pii}}}(\mathcal{N}_e(m))$
Check for existing enrolled accounts with u or `idx`; reject if found
Create account record using SEALACCOUNT
$E \leftarrow \text{Sign}_{k_{\text{sig}}}(\{\texttt{username} = u, \texttt{purpose} = \text{enroll}\}, \texttt{ttl} = 10 \text{ min})$
Return E to the client
Phase 2: Biometric Enrolment
User records V_r while speaking the displayed registration challenge
Client uploads V_r with enrolment token E
Verify E and extract identifier u'; reject if invalid or expired
Validate the upload and account; reject if invalid
Reject if a biometric template already exists for u'
$T_r \leftarrow$ EXTRACTTEMPLATE(V_r) with enrollment quality gate
if $T_r =$ **failure then**
    Delete temporary data and account record
    return rejection
end if
Generate the unique per-user biometric key $k_{u'}$
Store the protected $k_{u'}$ in the separate biometric key vault
Encrypt T_r using AES-256-GCM under $k_{u'}$
Store the encrypted template under $\mathcal{H}_{k_{\text{bio}}}(u')$
Destroy temporary biometric data
return registration successful

3.2 Login Phase

The login phase begins when the client submits the username to obtain a server generated, single use CAPTCHA. The user records a video while speaking the displayed challenge and submits the recording for authentication. The login recording is stored temporarily in encrypted form because it is consumed by both the biometric and challenge verification branches. The CAPTCHA is server side, associated with the authentication attempt, and consumed after use to prevent replay. The transient recording is removed once the authentication attempt is resolved.

Algorithm 2 Login and Challenge Issuance

Require: identifier u
Ensure: a fresh challenge and encrypted login recording
Validate identifier u; reject if invalid
$C \leftarrow$ randomly generate a five-symbol challenge
Store $(u, C, \texttt{expiry} = \text{now} + 5 \text{ min})$
Render and send the challenge image to the client
User records V_l while speaking C
Client uploads V_l
Validate the upload and identifier; reject if invalid
Verify that an unexpired challenge exists for u; reject otherwise
Store V_l in encrypted transient storage
Client requests authentication for u

3.3 Authentication Phase

The authentication phase operates on the enrolled biometric template and a newly submitted login recording. Raw enrolment recordings are not retained, while the login recording is stored only temporarily in encrypted form and deleted after the authentication attempt. The three factors are evaluated concurrently as follows.

1) The submitted identifier is checked against the user accounts.
2) The challenge issued for the identifier is retrieved and consumed, preventing its reuse in a subsequent authentication attempt regardless of the outcome.
3) The enrolled biometric template and encrypted transient login recording are decrypted into a private working directory. Failure to retrieve the user's key, decrypt, or authenticate either protected artefact rejects the attempt.
4) The login recording is processed once to obtain the login representations, consisting of per frame face embeddings and a speaker embedding.
5) The face and voice factors are scored against the enrolled template concurrently using (2) and (3).
6) Concurrently, the challenge branch extracts the login audio, transcribes it, and evaluates (4). This branch operates independently of the biometric matching branch.
7) The three verdicts are combined using the conjunctive rule in (5). On success, a signed session token is issued, on authentication failure, the response does not reveal which factor failed.
8) The private working directory and the transient encrypted login recording are removed regardless of the outcome.

3.4 Threat Model

3.4.0.1 Adversary capabilities.

The adversary is assumed to possess archived genuine face and voice media obtained from public sources or previous sessions, to control or modify the client device, to

TABLE 1: Challenge, grant, and account lifetimes. The implemented ordering is unenrolled account grace (15 min) > enrollment and re enrollment grant TTL (10 min) > login CAPTCHA TTL (5 min), so cleanup cannot remove an account while its enrollment grant remains valid.

Artefact	Lifetime	Use and single use constraint	Implementation source
Login CAPTCHA C	5 min	Single use; consumed on retrieval regardless of outcome	`captchaStore.js`
Enrollment token E	10 min	Bound to one account with purpose `enroll`; cannot replace an existing template	`enrollToken.js`, `server.js`
Re enrollment grant	10 min	Bound to one account with purpose `reenroll`; issued only after successful three factor authentication and may replace only an existing template	`enrollToken.js`, `appController.reenrollRequest`, `server.js`
Unenrolled account janitor grace	15 min	Rechecks template presence immediately before cleanup	`enrollmentJanitor.js`
Session token	Configuration dependent	Lifetime follows the signing environment configuration	`appController.js`

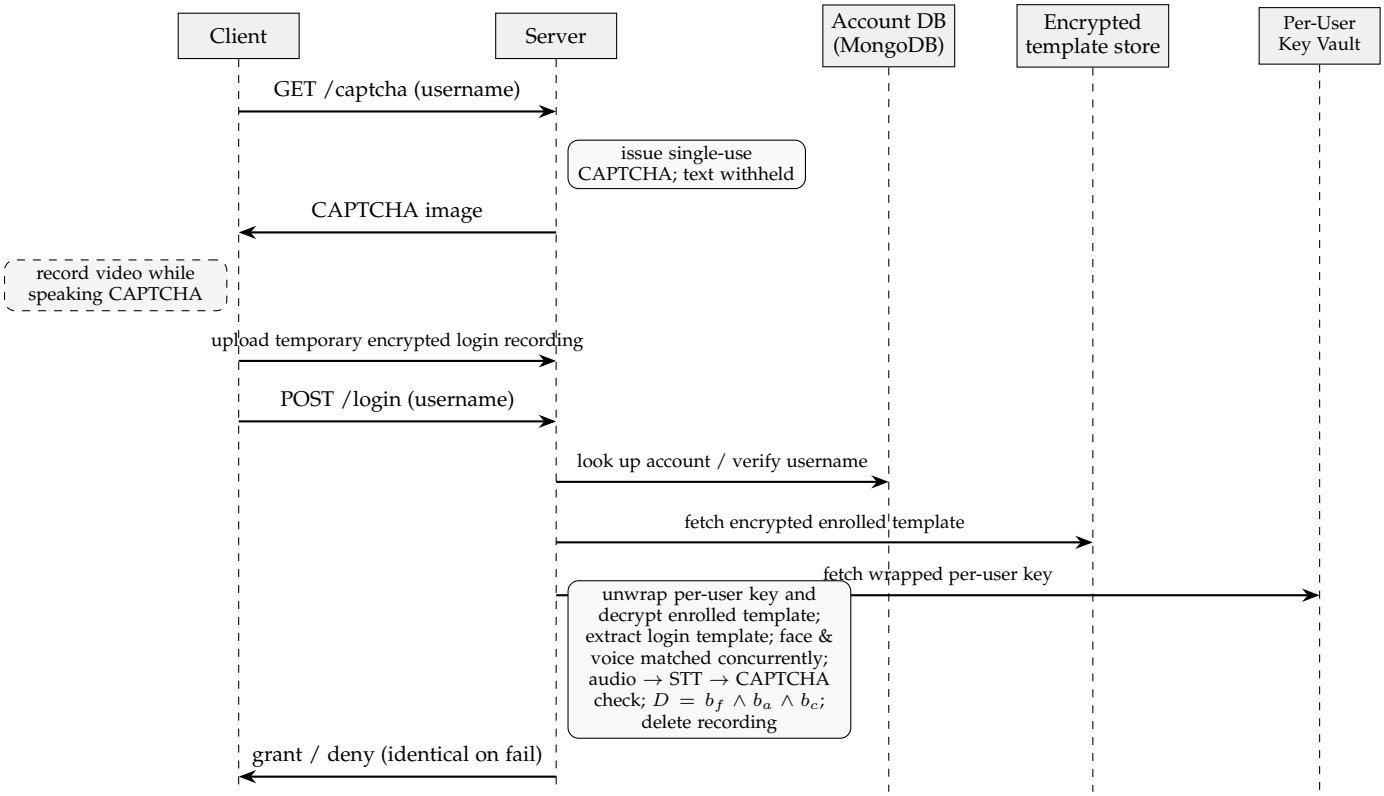


Fig. 2: Login and authentication phase (deployed)

observe or manipulate the network path, and to attempt phishing, relay, and replay based presentation attacks. Face and voice are observable biometric traits rather than secrets. Security therefore does not rely on biometric secrecy; it relies on a fresh server generated challenge, protection of stored templates and key material, and conjunctive verification of all three factors.

3.4.0.2 Trust assumptions.

The verifier, server side challenge generator, account store, template store, and separate key vault are trusted to execute the specified protocol and protect their server side secrets. The challenge is unpredictable, expires after five minutes, and is consumed regardless of the authentication outcome. The server is also trusted to enforce purpose bound enrollment and re enrollment grants and to return an identical client response for factor failures.

3.4.0.3 Out of scope.

A real time relay that presents the live challenge to the victim and returns the response within the validity window is out of scope. A real time responsive deepfake capable of generating the victim's face and voice while speaking an unpredictable challenge is also out of scope, as are verifier compromise, key vault compromise, coercion of the enrolled user, and operator abuse. The scheme therefore claims resistance to straightforward replay of previously captured media, but does not claim protection against these out of scope attacks.

3.5 System Model and Decision Rule

Let V_r and V_l denote the videos acquired during registration and login respectively, and let C denote the CAPTCHA

string issued by the server for the current session. Three decisions are derived from the single login stream:

- a face decision obtained by scoring the login recording's per frame face embeddings against the enrolled face embedding at a threshold τ_f;
- a voice decision obtained by scoring the login speaker embedding against the enrolled speaker embedding at a threshold τ_a;
- a challenge decision obtained by transcribing the login audio and comparing the transcript with C.

Both biometric factors are scored by *similarity* rather than distance. For an embedding $e \in \mathbb{R}^d$, the L2 (Euclidean) norm is

$$\|e\|_2 = \left(\sum_{j=1}^{d} e_j^2\right)^{1/2}, \qquad \widehat{e} = \frac{e}{\|e\|_2}. \tag{1}$$

L2 normalization therefore rescales an embedding to unit Euclidean norm. For two unit norm embeddings, $\langle \widehat{e}_1, \widehat{e}_2 \rangle = \cos\theta$, so their inner product is exactly their cosine similarity. Let e_r^f denote the normalized enrolled face embedding and $\{e_{l,i}^f\}_{i=1}^n$ the normalized per frame embeddings of the login recording, and let e_r^a and e_l^a denote the normalized enrolled and login speaker embeddings. Writing $T(\cdot)$ for the Speech to Text (STT) transform and $\mathcal{N}(\cdot)$ for the CAPTCHA normalization map, the factor decisions are

$$b_f = \mathbb{I}\Big[\operatorname{median}_i \langle e_r^f, e_{l,i}^f \rangle \;\geq\; \tau_f\Big], \tag{2}$$

$$b_a = \mathbb{I}\Big[\langle e_r^a, e_l^a \rangle \;\geq\; \tau_a\Big], \tag{3}$$

$$b_c = \mathbb{I}\Big[\exists\, t \in T(A_l) : \mathcal{N}(t) = \mathcal{N}(C)\Big], \tag{4}$$

where $\mathbb{I}[\cdot]$ is the indicator function. The face score is the *median* of the per frame similarities rather than the mean, so that a minority of degraded frames (motion blur, partial pose, transient occlusion) cannot move the verdict. The challenge decision accepts if *any* transcription hypothesis matches after normalization, because the speech to text stage returns a ranked list of alternatives and the correct reading of a spoken digit string is frequently not the top ranked one. Fusion is performed at the decision level under a conjunctive rule,

$$D = b_f \wedge b_a \wedge b_c, \tag{5}$$

A conjunctive rule is used so that compromise or successful spoofing of a single factor cannot independently produce a positive authentication decision. All three factors are evaluated concurrently and fused only after their verdicts are available. The rejection response is identical regardless of the failing factor, and per-factor outcomes are retained only in server side logs, reducing disclosure of factor specific information to the client.

Under an independence approximation, the conjunctive system satisfies

$$\mathrm{FAR}_{\mathrm{sys}} \approx \mathrm{FAR}_f \mathrm{FAR}_a \mathrm{FAR}_c, \tag{6}$$

$$\mathrm{FRR}_{\mathrm{sys}} \approx 1 - (1 - \mathrm{FRR}_f)(1 - \mathrm{FRR}_a)(1 - \mathrm{FRR}_c). \tag{7}$$

Thus AND fusion can lower the False Acceptance Rate while raising the False Rejection Rate relative to an individual factor. These expressions are structural approximations only; the present corpus is too small to instantiate them statistically.

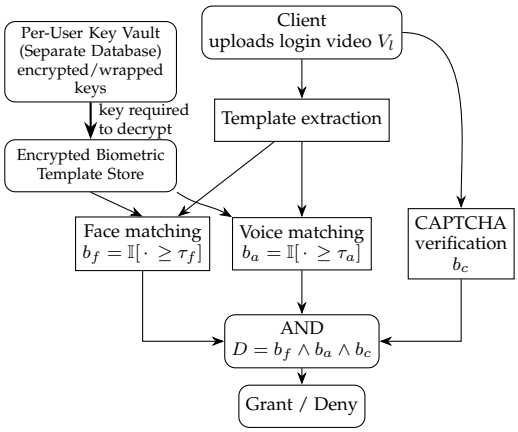


Fig. 3: Deployed concurrent authentication architecture. The login video is processed through concurrent face, voice, and CAPTCHA verification branches. Enrolled biometric templates are retained only in encrypted form, while the corresponding wrapped per user keys are maintained in a separate key vault. The three verdicts are combined by the conjunctive rule in (5), and access is granted only when all three pass.

3.6 Feature Representations

Face. The face verification factor uses a deep face recognition model to generate a 512 dimensional embedding for each detected face [13]. Twelve frames are sampled by evenly spaced frame index from each recording, and facial embeddings are extracted from the valid detected faces. The detector aligns and crops each face to the recognition model input. The enrolled reference is the L2 normalized mean of the frame embeddings. During authentication, the per frame face representations are compared with the enrolled reference using cosine similarity and the median score is used in (2). The face factor is accepted at $\tau_f = 0.35$. This value is an inherited operating point. The observed genuine score of 0.8385 and impostor score of 0.0928 support separation on the available recordings, but do not constitute an EER calibration.

Voice. The voice verification factor uses a 192 dimensional ECAPA TDNN speaker embedding [14]. Audio is resampled to 16 kHz mono before embedding extraction. The normalized speaker representation is compared with the enrolled reference using cosine similarity, as defined in (3). The voice factor is accepted at $\tau_a = 0.30$. This is an empirical operating point based on the available labelled observations: the maximum real cross speaker impostor cosine was approximately 0.256 and the genuine same speaker floor was approximately 0.345 after excluding one documented near silent four second window. The selected threshold lies in this observed gap. These diagnostic observations are not a formal FAR, FRR, or EER result. The superseded MFCC statistic produced a genuine range of 172.8 to 178.8 and an impostor range of 153.7 to 193.0, so the impostor range spanned the genuine range and did not provide a useful speaker identity threshold.

Acquisition variability. Device variation is partly absorbed by detector side face alignment, frame sampling by index rather than timestamp, median fusion of login face scores, cosine scoring of normalized embeddings, and audio conversion to 16 kHz mono. The browser requests ideal 1280 by 720 video at 30 frames per second and mono audio, with a one time fallback to unconstrained capture if negotiation fails. During enrollment only, a quality gate requires at least six usable sampled frames with a detected face of at least 80 by 80 pixels. The gate is not applied during login, where it could unnecessarily increase rejection of an already enrolled user. Per attempt width, height, frame rate, and platform metadata are retained in the benchmark log to support later cross device analysis. These mechanisms improve input consistency but do not establish cross device robustness, because no formal cross device experiment has yet been performed.

Once a sufficiently large consented corpus is available, each modality threshold will be swept over genuine pairs formed from different sessions of the same subject and impostor pairs formed from different subjects. The threshold at that modality's Equal Error Rate will be reported together with FAR and FRR at the deployed operating point. This calibration has not yet been performed.

3.7 Security Mechanisms

The proposed system incorporates multiple mechanisms to protect biometric data and the authentication workflow. During registration, the submitted video is processed to extract facial and voice biometric templates. The raw recording is not retained after successful processing, only the biometric representations required for subsequent verification are stored. This reduces the persistent exposure of reusable face and voice media and limits the amount of sensitive biometric data retained by the system.

Each enrolled user is assigned an independent cryptographically random 256 bit key that encrypts only that user's biometric template using AES-256-GCM. The per user key is not stored in plaintext with the biometric template or account data. Instead, it is stored in a separate key-vault database in encrypted form, where it is wrapped using a server side master key. The vault also uses a keyed pseudonymous index instead of storing usernames directly. Consequently, compromise of one user's template encryption key does not automatically expose the biometric templates of other users, and compromise of the template storage alone does not provide the corresponding decryption keys.

The system further protects the confidentiality and integrity of stored biometric data through authenticated encryption. During authentication, only the key associated with the requested user is retrieved and used to decrypt that user's template. The plaintext key exists only temporarily during the cryptographic operation and is cleared afterwards. Missing or tampered key material, as well as modified encrypted templates, cause the operation to fail rather than allowing authentication to continue.

The system also protects the association between an account and its biometric data during enrolment. Initial enrollment is authorized by a purpose bound `enroll` grant and is bound to the intended account through server controlled validation rather than client controlled information such as an uploaded filename. A normal enrollment grant cannot overwrite an existing enrolled template. Template replacement is available only through a purpose bound `reenroll` grant issued after successful three factor authentication, as described in Section 3.9.

During authentication, the server generates a dynamic challenge for each login attempt. The expected challenge value is not exposed to the client, is associated with the authentication attempt, and expires after a limited period. Authentication succeeds only when the face verification, speaker verification, and challenge response factors are all successful. Consequently, a previously recorded authentication attempt cannot independently satisfy a newly generated challenge, providing resistance against simple replay attacks.

Finally, user identifiers and uploaded inputs are validated before being used in processing or storage operations, reducing risks associated with untrusted filenames and path manipulation. Together, these mechanisms provide protection through template only storage, per user key isolation, separate key vault storage, authenticated encryption, integrity verification, secure enrolment, dynamic challenge response authentication, replay resistance, and input validation.

3.8 Comparison with Recent Authentication Systems (2020–2026)

This work presents an end to end multimodal biometric authentication system and therefore, the comparison is restricted to recent studies that explicitly implement a biometric authentication or verification procedure. Table 2 compares the proposed scheme with five relevant authentication systems published between 2024 and 2026 [15]–[19]. Works primarily focused on biometric recognition, presentation attack detection, cross-modal attacks, or template protection techniques [20]–[24] are considered supporting literature rather than directly comparable authentication systems. In Table 2, ✓ denotes a property that is supported or present, whereas ✗ denotes a property that is not supported, not implemented, or outside the scope of the corresponding system. For the proposed system, "challenge only" denotes challenge response evidence rather than a dedicated Presentation Attack Detection or deepfake detector.

Existing studies demonstrate the use of multimodal biometrics, liveness detection, challenge response mechanisms, and biometric template protection. The proposed work does not claim novelty for these individual components in isolation. Instead, its contribution lies in integrating facial verification, speaker verification, and a session specific spoken challenge within a single authentication architecture. The system further combines concurrent factor evaluation with strict decision level AND fusion, template only storage protected using authenticated encryption, pseudonymous template naming, enrollment binding, and explicit challenge lifecycle controls. As summarized in Table 2, the closest challenge response multimodal system [19] employs a spoken challenge for liveness, whereas the proposed architecture additionally incorporates server side single use challenge

management, independent per factor decisions combined through conjunctive fusion, protected template storage, and enrollment authorization.

The privacy and security advantage claimed in this comparison is the combination of protected templates, authenticated encryption, and unlinkability through pseudonymous naming. It is not a storage size comparison, because the compared works do not report comparable per user storage measurements. Raw enrollment and authentication recordings are not retained as permanent biometric records by the proposed system; for the compared systems, raw media retention is marked as not stated where the cited work does not specify it.

The proposed system concurrently evaluates the biometric and challenge response branches before producing a unified authentication decision. Since the current evaluation corpus is insufficient for meaningful FAR, FRR, or EER estimation, this work reports functional end to end authentication behavior, diagnostic matcher score separation, and measured processing overhead rather than cross system accuracy comparisons. Consistent with the threat model, the proposed challenge response mechanism mitigates the reuse of previously recorded authentication media but does not claim resistance to sophisticated real time relay attacks or real time deepfakes capable of responding to an unpredictable challenge within the validity window.

TABLE 2: Comparison of Security and Authentication Properties with Recent Systems (2024–2026)

Work	Liveness / PAD	Replay Prot.	Dyn. Chall.	Phish / MITM	Templ. Prot.	Encr.	Unlink./ Pseud.	Server dec.	Raw media permanently retained
Jha et al. [15]	✓	✗	✗	✗	✗	✗	✗	✗	not stated
Aanjankumar et al. [16]	✓	✓	✗	✓	✓	✓	✗	✗	not stated
Jha et al. [17]	✓	✗	✗	✗	✗	✗	✗	✗	not stated
Chen et al. [18]	✗	✗	✗	✗	✗	✗	✗	✗	not stated
Jha et al. [19]	✓	✓	✓	✗	✗	✗	✗	✓	not stated
Proposed Scheme	challenge only	✓	✓	✗	✓	✓	✓	✓	No, template only

3.9 Account Lifecycle

The deployed system implements the following account lifecycle.

- **Account creation and enrollment, implemented.** A purpose bound `enroll` grant authorizes initial template creation for the named account. Normal enrollment cannot overwrite an existing enrolled template. Username and email uniqueness are enforced, while biometric uniqueness across accounts is not.
- **Account update, implemented.** The authenticated `PUT /api/updateuser` operation updates account information. Changed personally identifiable information is sealed again, and a changed email address receives a new keyed index after collision checking.
- **Authenticated re enrollment, implemented.** `POST /api/reenroll-request` is answered only after the same successful face, voice, and challenge verification used for login. The resulting purpose bound `reenroll` grant is valid for ten minutes and may replace only an existing biometric template. Replacement uses a newly generated per user key and an atomic ciphertext swap; failure before completion preserves the prior enrollment.
- **Account deletion and biometric erasure, implemented.** The authenticated `DELETE /api/account` operation removes the account record, encrypted biometric template, separate vault key entry, and any transient login recording associated with the account.
- **Operator recovery, administrative procedure.** If a user cannot complete three factor authentication, recovery is not exposed as a public API endpoint. An operator verifies identity out of band, deletes the existing account through the administrative equivalent of the erasure procedure, and permits the user to register and enroll again. This administrative process is the explicit recovery trust anchor.

4 Evaluation and Results

4.1 Experimental Setup

A proof of concept web application was implemented to validate the feasibility of the proposed scheme under real time acquisition. The prototype follows a client–server architecture in which the client performs media acquisition and displays the server issued challenge, while enrolment processing, challenge generation, feature extraction, matching, and decision fusion are performed on the server. Keeping the decision logic server side prevents a modified client from directly bypassing authentication checks.

4.1.1 Hardware Requirements

Functional verification was performed on a machine equipped with an Intel Core i5-4210U processor and 8 GB of RAM. The processing-time measurements reported in Section 4.2 were collected on a separate machine equipped with an Intel Core i9-12900 processor and 62 GB of RAM. The reported timing results should therefore be interpreted with respect to the second machine.

4.1.2 Prototype Architecture

The implementation uses a React single page client, an Express server on Node.js 22, MongoDB for account and storage data, and Python 3.12 for the biometric pipeline. Face inference is performed through ONNX Runtime, while speaker processing uses PyTorch. Production deployments require independent secrets for token signing, biometric storage, and account/PII protection. The server validates these secrets at startup and rejects missing, insecure, or improperly shared production configuration.

4.2 Processing and Storage Footprint

Controlled per stage processing time was measured over $N = 5$ successful genuine authentication runs on the timing machine described in Section 4.1.1. Additional live deployment measurements are reported separately in Table 3 using the sample counts shown in the table.

Login template extraction is the dominant processing stage because it loads and executes both the face and speaker model stacks. Model loading contributes substantially to this overhead. By comparison, face and voice matching require only similarity computations between extracted and enrolled representations and introduce comparatively low overhead.

Storage footprint. The persistent per user biometric artefact is approximately 147 KB, while the source recording used for this measurement was 11.18 MB, an approximately 76 fold difference for that recording. The ratio depends on recording duration, resolution, frame rate, and codec and is therefore reported only as a deployment storage measurement, not as a generic security, accuracy, or comparative advantage. Template storage is $O(1)$ with respect to recording length because the stored representation has fixed dimensions: twelve 512 dimensional face embeddings, one 512 dimensional mean face reference, and one 192 dimensional speaker embedding. The AES 256 GCM envelope adds a fixed 34 byte overhead. Network upload time was not measured and is therefore not reported as an experimental result.

The implemented recognition evaluation harness comprises `evaluate_corpus.py`, the `run_evaluation.mjs` driver, and the collection and execution protocol in `docs/EVALUATION.md`. It applies the deployed face and voice matcher paths to a labelled corpus, forms genuine pairs only between different sessions of the same subject, forms impostor pairs between different subjects, and can produce score distributions, Receiver Operating Characteristic (ROC) curves, Detection Error Tradeoff (DET) plots, Area Under the Curve (AUC), and Equal Error Rate estimates. The harness is ready for a sufficiently large, separately consented corpus, but such a corpus has not yet been collected and the harness has therefore not produced reportable recognition accuracy results.

For a score threshold τ, the False Acceptance Rate is $\mathrm{FAR}(\tau) = P(s \geq \tau \mid \text{impostor})$, and the False Rejection Rate is $\mathrm{FRR}(\tau) = P(s < \tau \mid \text{genuine})$. The Equal Error Rate is the error value at the threshold where FAR and FRR are equal. These definitions describe the planned evaluation and are not measurements from the present recordings.

Table 4 reports diagnostic genuine and impostor scores for the deployed face and voice matchers. These examples support operating point discussion, but they are not a recognition accuracy evaluation and no FAR, FRR, EER, ROC, DET, or AUC value is claimed from them.

Tables 5 and 6 report the end to end workflow timing and operational outcomes from the development deployment, while Table 7 summarizes the diagnostic evidence used for the voice operating point.

4.3 Evaluation Scope and Limitations

The evaluation measured processing time, storage footprint, diagnostic matcher score behaviour, and the end to end functional behaviour of the deployed authentication workflow.

No formal cross device evaluation has yet been performed. Although the capture client requests ideal 720p video at 30 frames per second with mono audio and records negotiated device metadata, all current measurements should be interpreted as same environment deployment observations rather than evidence of cross device robustness.

No formal recognition accuracy evaluation was performed. The available corpus is too small to support meaningful estimates of False Acceptance Rate, False Rejection Rate, Equal Error Rate, or ROC based performance. The deployed face and voice thresholds are operating points rather than calibrations against a target error rate.

Enrollment quality is now gated by minimum face size and a minimum number of usable sampled frames. However, template only storage means that discarded enrollment recordings cannot be retrospectively recomputed or re scored. Future recognition accuracy analysis therefore requires a separately collected, consented, multi subject corpus with multiple sessions and devices. The existing evaluation harness can process such a corpus when it becomes available.

The planned recognition evaluation fields and pairing protocol are summarized in Table 8. No values are reported because no sufficiently large consented corpus has yet been collected.

TABLE 3: Measured per stage processing time (ms). Controlled rows use five successful genuine authentication runs on the i9-12900 timing machine. Live rows use the deployed server benchmark log for four successful logins from the 25 attempts recorded on August 22, 2026; they include ordinary system contention. Both sets show that template extraction dominates and matching contributes comparatively little latency.

Stage	Mean	Min	Max	N	Source
Login template extraction	12424	11888	12779	5	Controlled runs, i9-12900
Face matching	98	73	116	5	Controlled runs
Voice matching	104	83	138	5	Controlled runs
Audio demultiplexing	154	146	162	5	Controlled runs
Speech to text	2367	2085	2666	5	Controlled runs
Template decryption	**9.4**	**2.6**	**28.8**	**4**	**Live log, successful logins**
Login template extraction, live	**6596**	**4606**	**12190**	**4**	**Live log, successful logins**
Face and voice matching, combined, live	**150**	**143**	**155**	**4**	**Live log, successful logins**
Speech to text, live	**2354**	**1534**	**3648**	**4**	**Live log, successful logins**
Server side verification, total, live	**6746**	**4757**	**12339**	**4**	**Live log, successful logins**

TABLE 4: Deployed matcher diagnostic scores and threshold provenance. The observations support operating point selection but are not a formal FAR, FRR, or EER evaluation.

Deployed matcher	Genuine	Impostor	Threshold	Threshold provenance
Face: median per-frame cosine	0.8385	0.0928	$\tau_f = 0.35$	Inherited operating point; observed separation supports its use on the available recordings, but it is not EER calibrated
Voice: ECAPA-TDNN speaker cosine	0.455	0.090	$\tau_a = 0.30$	Empirical operating point; the observed impostor maximum is 0.256 and genuine floor is 0.345, so 0.30 lies in the measured gap

TABLE 5: End to end authentication and enrollment time from the deployed development log ($N = 25$ attempts). Client pre-upload time includes recording the video and speaking the challenge. For successful login, this user time is approximately 20 seconds of the approximately 27 second total, while mean server processing is 6.76 seconds.

Operation	Outcome	N	Total mean (s)	Total min (s)	Total max (s)	Client pre-upload mean (s)	Server mean (s)
Registration, account and enrollment	success	7	11.56	7.35	14.30	1.55	10.01
Login	success	4	26.89	14.87	42.78	20.13	6.76
Login	failure	13*	35.08	0.02	140.08	21.13	12.95

*The log contains 14 failed login records; one has no total because it was rejected before any upload.

TABLE 6: Operational outcomes from the development deployment ($N = 25$ attempts). These are workflow outcomes, not biometric error rates. No logged failure was a biometric mismatch. `NO_ENROLLMENT` records describe the abandoned enrollment condition addressed by the enrollment janitor and pending enrollment state; `CAPTCHA_MISMATCH` and `SPEECH_NOT_RECOGNIZED` are third factor transcription failures. Neither FAR nor FRR can be inferred from this table.

Operation	Outcome	Reason code	Count
Registration	success	not applicable	7
Login	success	not applicable	4
Login	failure	`NO_ENROLLMENT`	7
Login	failure	`CAPTCHA_MISMATCH`	6
Login	failure	`SPEECH_NOT_RECOGNIZED`	1
Total			**25**

TABLE 7: Voice threshold derivation evidence from the available diagnostic observations. The zero measured false acceptances apply only to the available real impostor pairs and are not a formal corpus-level FAR estimate.

Statistic	**Value**	**Source**
Real cross-speaker impostor cosine, maximum	0.256	`matchVoiceTemplate.py:24--40`; `docs/voice-verification.md`
Genuine same-speaker cosine, floor	0.345, excluding one near-silent 4 second window	Same diagnostic source
Selected operating point τ_a	0.30	`matchVoiceTemplate.py:50`
Measured FAR on the available real impostor pairs at τ_a	0; diagnostic observation only, not a formal corpus-level FAR estimate	Same diagnostic source
Superseded MFCC statistic, genuine range	172.8 to 178.8	`extractTemplate.py:17--21`
Superseded MFCC statistic, impostor range	153.7 to 193.0; spans the genuine range	Same diagnostic source

TABLE 8: Planned recognition accuracy evaluation. No sufficiently large consented corpus has yet been collected, so the fields are not available and are not experimental results. Genuine pairs will be formed only between different sessions of the same subject, while impostor pairs will be formed between different subjects.

Modality	**Genuine pairs**	**Impostor pairs**	**EER (%)**	**AUC**	**FAR at deployed τ (%)**	**FRR at deployed τ (%)**	**τ at EER**
Face, median per-frame cosine	not available	not available	not available	not available	not available	not available	not available
Voice, ECAPA-TDNN cosine	not available	not available	not available	not available	not available	not available	not available
Challenge, STT versus C	not available	not available	not applicable	not applicable	not available	not available	not applicable
Fused, $b_f \wedge b_a \wedge b_c$	not available	not available	not available	not available	not available	not available	not applicable

Key rotation is not yet automated and remains an operational limitation. The system separates signing, biometric storage, and account data protection keys, but replacing a storage key currently requires controlled migration of the corresponding protected data. Future work should provide automated key versioning and online re-encryption to support rotation without service interruption.

5 Future Work

A primary direction for future work is to strengthen the proposed authentication system against deepfake and advanced presentation attacks. Although the dynamic CAPTCHA provides challenge response based protection against simple replay attacks, an attacker may potentially use AI generated or manipulated facial and voice content to imitate an enrolled user. Therefore, future work will investigate the integration of dedicated deepfake and presentation attack detection mechanisms for both facial and voice modalities.

For the facial modality, future work can incorporate face anti spoofing and deepfake detection techniques to distinguish genuine live facial input from replayed, manipulated, or synthetically generated video. Similarly, for the voice modality, synthetic speech and voice cloning detection can be integrated to identify AI generated or converted speech. These additional checks can operate alongside the existing face verification, speaker verification, and challenge response mechanisms, providing an additional layer of protection rather than replacing the current authentication factors.

Future work will also focus on evaluating the complete system against a broader range of attack scenarios involving manipulated video, cloned speech, and multimodal deepfakes. The goal will be to determine whether face, voice, and liveness related evidence can be jointly analysed to improve resistance against attacks in which multiple biometric modalities are synthetically generated or manipulated simultaneously.

In addition, a larger multi subject evaluation corpus will be collected to measure recognition performance using metrics such as false acceptance rate, false rejection rate, and equal error rate. Further work will also investigate reducing the overhead of template extraction through persistent model services and replacing the external speech to text dependency with an on premises solution to improve privacy and reduce reliance on third party services.

A cross device protocol will record each consenting subject in multiple sessions on at least two devices and will report within device and cross device genuine score distributions, impostor distributions, FAR, FRR, EER, ROC, DET, and AUC results produced by the existing evaluation harness. This evaluation is future work and has not yet been run. Automated key rotation and online re encryption also remain future operational work. No dedicated facial or voice deepfake detector, new Presentation Attack Detection model, or automated key rotation mechanism is claimed as part of the current implementation.

6 Conclusion

This paper presents a single sensor multimodal biometric authentication system that combines face verification, speaker verification, and a dynamic challenge response mechanism using a single real time video recording. Facial and voice information are acquired simultaneously, while the user responds to a server generated CAPTCHA. Authentication is granted only when all three factors are successfully verified through a conjunctive decision rule.

The implemented system uses deep face embeddings and cosine similarity for face verification, ECAPA-TDNN speaker embeddings and cosine similarity for speaker verification, and speech to text based challenge validation.

Raw enrollment recordings are deleted after template extraction, while encrypted login recordings are retained only transiently and deleted when the authentication attempt resolves. The resulting biometric templates are protected and stored separately from account information, supporting a privacy conscious authentication workflow.

The evaluation measured controlled processing time on the timing machine and reported additional end to end workflow observations from the deployed development system, including registration, successful login, and failed login attempts. Template extraction remained the dominant processing stage, while face and voice matching introduced comparatively low overhead. Recognition accuracy was not statistically characterised because the available corpus was too small for meaningful estimates of metrics such as false acceptance and false rejection rates.

Overall, the proposed system demonstrates a deployable and practical multimodal authentication architecture using commonly available camera and microphone hardware while reducing persistent storage of raw biometric media. It provides a foundation for more robust and privacy conscious multimodal authentication systems.

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